

Week 5 - Atmospheric Dynamics: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS**: selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY**: expected independent practice after the session.
- **EXTRA CHALLENGE**: optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Useful constants: Earth rotation rate $\Omega = 7.292 \times 10^{-5} \text{ s}^{-1}$. The Coriolis parameter is $f = 2\Omega \sin \phi$. Horizontal pressure-gradient acceleration may be estimated as $a_p = -(1/\rho) \Delta p / \Delta x$.

YOU SHOULD TRY Question 1: Identify the force balance from the situation

Complete the table by naming the dominant balance or additional force in each case. Then give one reason why the approximation might fail.

Situation	Dominant balance / One limitation force
Vertical atmosphere outside a strong convective updraught	
Straight upper-level flow at 50°N, well above surface friction	
Wind at 10 m over rough land	
Flow very close to the equator	

YOU SHOULD TRY Question 2: Horizontal pressure-gradient acceleration

Sea-level pressure is 1014 hPa to the south of a region and 1008 hPa to the north, separated by 300 km. Take air density to be 1.20 kg m^{-3} .

1. Calculate the magnitude of the horizontal pressure-gradient acceleration.
2. State its direction.
3. If the same 6 hPa pressure change occurred across only 150 km, by what factor would the acceleration change?

YOU SHOULD TRY Question 3: Coriolis acceleration over Manchester

At latitude 53.5°N, calculate the Coriolis parameter f and the magnitude of the Coriolis acceleration for a wind speed of 15 m s^{-1} .

1. What is the magnitude of f ?
2. What is the acceleration magnitude?
3. If the wind is blowing due east, in which compass direction does the Coriolis acceleration point?

IN CLASS Question 4: Geostrophic wind speed and direction

Use the pressure gradient from Question 2 and the latitude from Question 3. Assume straight isobars and negligible friction.

1. Estimate the geostrophic wind speed.
2. Determine whether the geostrophic wind is approximately eastward or westward.
3. State where the lower pressure lies relative to the geostrophic wind in the Northern Hemisphere.

EXTRA CHALLENGE Question 5: Use isobar spacing to compare winds

Region A has isobars every 4 hPa separated by 80 km. Region B has the same 4 hPa interval separated by 200 km. Assume the same latitude and density.

1. Calculate the ratio of the pressure-gradient magnitudes, $|\nabla p|_A/|\nabla p|_B$.
2. Hence estimate the ratio of geostrophic wind speeds.
3. If the geostrophic wind in Region B is 12 m s^{-1} , estimate it in Region A.
4. Explain why the actual 10-m wind in either region would usually be weaker and would cross the isobars toward lower pressure.

YOU SHOULD TRY Question 6: Diagnose a sea-breeze circulation from pressure data

During a sunny afternoon, measurements give the following idealised pressures:

	Over land	Over sea
Surface	1007 hPa	1010 hPa
At 1.5 km	850 hPa	846 hPa

1. At the surface, in which direction does the pressure-gradient acceleration act?
2. At 1.5 km, in which direction does it act?
3. Use these two answers to sketch the circulation in a vertical land-sea cross-section, including rising and sinking branches.
4. Why is this circulation normally strongest after several hours of daytime heating rather than just after sunrise?

YOU SHOULD TRY Question 7: From a surface pressure system to vertical motion and weather

Two idealised Northern Hemisphere pressure systems are observed.

System	Near-surface wind	Central pressure tendency
A	spirals inward anticlockwise	low pressure
B	spirals outward clockwise	high pressure

For each system, state whether the surface flow converges or diverges, infer the sign of the vertical motion required by mass continuity, and predict the broad cloud/precipitation tendency if moisture is available.

EXTRA CHALLENGE Question 8: Use the three-cell circulation to diagnose latitude bands

For each latitude band in the table, fill in the most likely broad vertical motion and the prevailing surface-wind family in the Northern Hemisphere.

Latitude	Vertical motion	Prevailing surface wind
0°	?	?
30°N	?	?
50°N	?	?
60°N	?	?
90°N	?	?

Then identify which of these latitude bands is most strongly associated with (i) persistent tropical rainfall and (ii) subtropical desert belts.

EXTRA CHALLENGE Question 9: Temperature contrast and the jet stream

Two winter air columns have the same surface pressure. At a fixed pressure level aloft, the surface-to-level thickness is 5.7 km in the warm column but 5.2 km in the cold column.

1. Which column places that pressure surface at the greater geometric height?
2. What does the 0.5 km height difference imply about the horizontal pressure-gradient tendency aloft across the boundary between the columns?
3. Use geostrophic reasoning to explain why a strong upper-level wind tends to occur near a strong horizontal temperature contrast, and why the mid-latitude jet is often stronger in winter.

Modelling connection.

These harder dynamics problems show how geostrophic balance, curved flow and Rossby-wave ideas appear in simple atmospheric models. Students who also take EART22001 will meet related ideas in a shallow-water model, but no knowledge of that unit is assumed here.

EXTRA CHALLENGE Question 10: Gradient wind around a curved low-pressure system

At latitude 50°N, an air parcel follows circular isobars around a low with radius of curvature $R = 500$ km. The inward pressure-gradient acceleration is $a_p = 2.0 \times 10^{-3} \text{ m s}^{-2}$. Neglect friction. For cyclonic flow in the Northern Hemisphere use the radial balance

$$\frac{v^2}{R} + fv = a_p, \quad f = 2\Omega \sin \phi.$$

1. Calculate f .
2. Solve the quadratic for the physically relevant gradient-wind speed v .
3. Calculate the geostrophic speed $v_g = a_p/f$ for the same pressure gradient.
4. Is the cyclonic gradient wind subgeostrophic or supergeostrophic here? Explain using the force balance.

EXTRA CHALLENGE Question 11: Thermal wind: why jets strengthen with horizontal temperature contrast

At 50°N, the mean temperature at 850 hPa decreases northward by 8 K over 1000 km. Use the pressure-coordinate thermal-wind estimate

$$\Delta u_g \approx -\frac{R_d}{f} \frac{\partial T}{\partial y} \ln \left(\frac{p_2}{p_1} \right),$$

with $p_1 = 850$ hPa, $p_2 = 300$ hPa and $f = 1.12 \times 10^{-4} \text{ s}^{-1}$.

1. Estimate $\partial T / \partial y$ in K m^{-1} .
2. Estimate the change in geostrophic zonal wind between 850 and 300 hPa.
3. Explain why a stronger equator-to-pole temperature gradient in winter is associated with a stronger mid-latitude jet.

IN CLASS Question 12: Orographic Rossby-wave reasoning

A barotropic westerly flow crosses an isolated north-south mountain ridge. The flow is displaced poleward on one side of the disturbance and equatorward on the other. Assume the Coriolis parameter increases with latitude.

1. If a parcel moves poleward while conserving absolute vorticity approximately, how must its relative vorticity change?
2. What happens for an equatorward displacement?
3. Explain qualitatively how alternating meridional displacements can support a Rossby-wave pattern downstream of topography.
4. Why is geostrophic balance a useful starting point for initialising such a shallow-water experiment?